

INVESTMENT ADVISORY MEMORANDUM

TO: Portfolio Manager
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SUBJ: Strategy for Client Equity Exposure

1 Client Situation

The client holds a high conviction position of **500 shares** of a large cap stock trading at a price (S_0) of **2,450**, representing a value of **12,25,000**.

The client eyes a **10% to 15% market correction** over a **3 month horizon** ($T = 0.25$ years). Physical liquidation of the asset is prohibited due to:

- **Tax Friction:** Selling triggers immediate Short Term or Long Term Capital Gains Tax liabilities under income tax regulations.
- **Conviction:** The client maintains a long term bullish outlook beyond the 3 month window.

2 Call vs Put Options

To mitigate downside risk without triggering share liquidation, we analyze the terminal payoff profiles of options at maturity (T).

2.1 The Downside Shield: Long Put Option

A Put Option grants the holder the right, but not the obligation, to sell the asset at a fixed strike price (K). Its terminal payoff is:

$$\text{Payoff}_{\text{Put}} = \max(K - S_T, 0) \quad (1)$$

If S_T drops below K , the payoff increases linearly, creating an exact mathematical counterweight to the losses sustained in the physical stock portfolio.

2.2 The Discrepancy: Call Option

A Call Option grants the right to buy an asset at strike K . Its terminal payoff is:

$$\text{Payoff}_{\text{Call}} = \max(S_T - K, 0) \quad (2)$$

Calls require a bullish outlook. In a market correction where $S_T \leq K$, a call option expires completely worthless (Payoff = 0), providing zero hedging capability.

Numerical Example: Let $K = 2,400$. If a correction drives the stock price down to $S_T = 2,100$:

- $\text{Payoff}_{\text{Put}} = \max(2400 - 2100, 0) = \mathbf{300}$ per share (Offsets portfolio loss by yielding 1,50,000 across 500 shares).
- $\text{Payoff}_{\text{Call}} = \max(2100 - 2400, 0) = \mathbf{0}$ (Fails as a hedge).

So, clearly Put option is the way for us, here.

3 Strike Selection & Moneyness

We propose purchasing an **Out-of-the-Money (OTM) Put Option** with a strike price of $K = 2,350$.

- **Strategic Rationale:** An At-the-Money (ATM) Put ($K = 2,450$) offers immediate downside protection but carries an expensive time-value premium due to maximum uncertainty. The OTM Put ($K = 2,350$) introduces a $100/2450 * 100 \sim 4\%$ risk buffer (deductible) but drastically lowers the upfront premium cost, optimizing capital efficiency.
- **Premium Deconstruction:** At the current spot ($S_0 = 2,450$), the contract is **OTM** since $S > K$.

$$\text{Option Price} = \text{Intrinsic Value} [\max(2350 - 2450, 0) = 0] + \text{Time Value} \quad (3)$$

The entire premium consists solely of *Time Value*, tracking the probability of the stock falling below 2,350 before expiry.

4 European vs. American

- **European Options:** Exercisable strictly at maturity (T).
- **American Options:** Exercisable at any point up to and including maturity (T).

For an American Put, **early exercise can become optimal** if the stock price drops sharply deep In-the-Money ($S \ll K$). In this scenario, the put's remaining time value collapses to zero. Under positive risk-free interest rates ($r > 0$), the holder benefits by exercising immediately to capture the cash value ($K - S$) and investing it at rate r to earn continuous interest, rather than waiting until maturity. This makes pricing the American put an *optimal stopping free boundary problem*.

5 Payoff Matrix

Assuming a market option premium of **50 per share** (Total Premium Outlay = 25,000 for 500 shares), the consolidated position behaves as follows at expiry:

Spot Price (S_T)	500 Shares Value	Put Payoff / Share	Total Put Payoff	Net Portfolio Value	Net Strategy P&L
2,100 (-14.3%)	10,50,000	250.00	1,25,000	11,75,000	-50,000 (Capped)
2,300 (-6.1%)	11,50,000	50.00	25,000	11,75,000	-50,000 (Capped)
2,450 (Flat)	12,25,000	0.00	0	12,25,000	-25,000 (Premium)
2,600 (+6.1%)	13,00,000	0.00	0	13,00,000	+50,000
2,700 (+10.2%)	13,50,000	0.00	0	13,50,000	+1,00,000

Breakeven Math: Standalone option breakeven is $K - \text{Premium} = 2,300$. Integrated portfolio breakeven is $S_0 + \text{Premium} = \mathbf{2,500}$ (requiring a 2% stock appreciation to cover the insurance cost).

6 Trade Recommendation

- **Execution Strategy:** I recommend implementing a formal **Protective Put Strategy** by purchasing 500 units of individual stock OTM Put Options at the 2,350 strike (3 Month Maturity). Total budget allocation is **25,000** ($\sim 2.04\%$ of the asset base). If specific stock contracts face liquidity constraints on the National Stock Exchange (NSE), an equivalent proxy hedge should be executed via liquid **Nifty 50 Index Put Options**, scaled according to the stock's dynamic characteristics.
- **Primary Operational Risks:**
 1. *Theta Decay:* Options are wasting assets; time decay accelerates non-linearly in the final 30 days before expiration.
 2. *Liquidity Risk:* Individual NSE equity options can exhibit wide bid-ask spreads, increasing the cost of exiting positions early.
 3. *Basis Risk:* If an index proxy option is used, company-specific stock drops might not be covered if the wider index remains flat or rises.

7 Analytical Reflection

Engaging with the interactive payoff explorer refined my perspective on option mechanics. I originally viewed option prices as simple static variables. Seeing the "kink" at $S = K$ mapped out that this transition is a non-differentiable boundary condition. This explains why standard closed-form calculus equations like Black-Scholes cannot solve American options, turning them into dynamic free-boundary optimization problems. **References:**

1. Investopedia, *Protective Put Strategy Guide Framework*. URL: <https://www.investopedia.com/terms/p/protective-put.asp>
2. National Stock Exchange of India (NSE), *Equity Derivatives Specifications Framework*.